

NARI NIKETAN

COMPLETE SYSTEM ARCHITECTURE & ENGINEERING SPECIFICATION

Official Technical Architecture Blueprint & System Manual

Version: 3.0.0 (Production Node.js Backend & Multi-Portal Ecosystem Release) | Date: September 2026
Domain: <https://www.nariniketan.shop> | Firebase Hosting: <https://nari-niketan.web.app>
Backend API (Cloud Run): <https://nari-niketan-api-997712460310.asia-south1.run.app>

System Dimension / Pillar	Architectural Specification & Implementation
Architecture Pattern	Hybrid Jamstack + Microservice API on Google Cloud Run (asia-south1, Mumbai)
Frontend Presentation Tier	Progressive Web App (PWA), Semantic HTML5, CSS3 Custom Tokens, Vanilla ES6+ JS
Backend API Layer	Node.js 20 LTS, Express 4.19, Helmet, CORS Allowlist, Rate Limiters, Winston Audit Logging
Cloud Database & Storage	Cloud Firestore NoSQL (17 collections, atomic transactions) + Firebase Storage (5MB MIME)
Authentication & RBAC	Firebase Auth + Custom Claims (admin, seller, owner) + Firestore Security Rules
Pricing & Discount Authority	100% Server-Authoritative: Zero Trust on client-sent prices, totals, or discount rates
Multi-Portal Ecosystem	4 Unified Portals: Customer Storefront, Admin Operations, Vendor/Seller, Delivery Partner
Logistics & Fleet Engine	Zone matching, OTP handoff verification, GPS Google Maps navigation, COD reconciliations
Payment Settlement	Dynamic BharatPe UPI VPA QR generator with UTR manual audit + Cash on Delivery (COD)
Edge Caching & Offline	Service Worker (v6.0) with offline precache shell + dynamic bypass for real-time portals

Document Classification: OFFICIAL ENGINEERING SPECIFICATION & PRODUCTION BLUEPRINT

Section 1 — Multi-Tier System Topology & Security Boundary

The Nari Niketan platform is organized into three distinct, decoupled architectural layers designed for sub-second load times, defense-in-depth security, and infinite auto-scaling:

- **1. Client Presentation Tier (Edge CDN):** Hosted across Google Edge CDN / Firebase Hosting with HTTP/2 and TLS 1.3. Serves the responsive customer storefront, PWA shell, and management portals. Contains zero business secrets.
- **2. Application Security & Logic Tier (Cloud Run):** Containerized Node.js/Express service deployed in asia-south1 (Mumbai). Validates all incoming payloads with Zod schemas, fetches ground-truth catalog prices from Firestore, enforces coupon quotas, and signs state changes.
- **3. Cloud Persistence & Identity Tier (Google Cloud / Firebase):** Cloud Firestore manages NoSQL documents with atomic batch transactions. Firebase Auth issues cryptographically signed JWT ID tokens containing role claims. Cloud Storage stores product photography.

Section 2 — Multi-Portal Ecosystem & Access Control Matrix

Portal	Target User	Primary Capabilities	Auth & RBAC Control
Storefront PWA /index.html	Shoppers / Customers	Catalog browsing, faceted filters, AI Stylist advisor, server-validated cart, UPI/COD checkout, order tracking.	Public / Authenticated Customer JWT
Admin Operations /admin	Super Admins & Ops	Catalog management, stock editing, dispatch assignment, coupon creation, seller verification, finance payouts.	Firebase Custom Claim admin: true or owner: true
Seller Portal /seller	Artisans & Vendors	Vendor onboarding, product creation/editing, inventory sync, vendor order inspection, and payout settlements.	Firestore status sellerStatus: 'approved'
Delivery Partner /delivery	Fleet Riders & Agents	Shift availability, assigned delivery queue, Google Maps navigation, customer dialing, and OTP handoff.	Firestore status deliveryPartnerStatus: 'approved'

Section 3 — Backend Microservice Design & Server Authority

The backend API service (nari-niketan-api) is containerized via Docker and runs on Google Cloud Run in the asia-south1 region with automated auto-scaling and IAM authorization.

Component / Service	Implementation Details & Engineering Rules
Pricing Service pricingService.js	Computes line items, category subtotals, dynamic free shipping thresholds (Rs. 999 threshold / Rs. 99 standard fee), and 0% GST (inclusive). Ignores all client-submitted totals.
Coupon Engine couponService.js	Validates coupon active state, expiration dates, minimum cart spend, per-user usage limits, and computes percentage vs flat discounts strictly on the server.
Order Processor orderService.js	Fetches active product documents directly from Firestore, checks inventory levels, generates 6-digit delivery OTPs, records coupon usage, and writes orders atomically.
Input Validation orderValidator.js	Strict Zod schema validation for cart payloads, addresses (6-digit PIN codes, line items), and phone numbers. Rejects invalid requests with HTTP 400 before DB access.
Security Middleware helmet & cors	Enforces strict Content Security Policy, HSTS, X-Frame-Options DENY, and CORS origin whitelist restricted to nariniketan.shop and Firebase domains.
DDoS Rate Limiting rateLimiter.js	Tiered memory rate limiters: General endpoints (100 req/15 min), Order placement (10 req/15 min), Coupon validation (20 req/15 min).
Structured Logging logger.js	Winston logger outputting structured JSON logs for Google Cloud Logging, capturing request IDs, audit trails, and execution latencies.

Section 4 — Cloud Firestore NoSQL Data Architecture

Collection Name	Key Schema Fields & Data Types	Security & Read/Write Access
/users/{uid}	email, displayName, role, isAdmin, isSeller, sellerStatus, isDeliveryPartner, deliveryPartnerStatus, shiftStatus, deliveryProfile	Owner read/write profile; Admin manages roles & statuses.
/products/{id}	title, price, originalPrice, stock, category, sellerId, images[], status, tags[], rating	Public read; Admin & owning Seller create/update/delete.
/orders/{id}	userId, customerName, phone, deliveryAddress{}, items[], subtotal, discountAmount, shipping, totalAmount, paymentMethod, paymentStatus, upiUtr, deliveryOtp, status, assignedDeliveryPartnerId, deliveryState	Admin all; User reads own; Seller reads items; Assigned Rider reads and updates delivery status.
/coupons/{id}	code, discountType, discountValue, minCartAmount, maxDiscount, validUntil, usageLimit, usedCount, active	Admin manages; Authenticated users validated via Cloud Run.
/deliveryPayouts/{id}	deliveryPartnerId, amount, orderIds[], status, processedAt, notes	Admin manages; Rider reads own payout history.
/sellerPayouts/{id}	sellerId, amount, orderIds[], status, processedAt, notes	Admin manages; Seller reads own settlements.
/reviews/{id}	productId, userId, userName, rating, comment, createdAt	Public read; Authenticated verified buyer create.

Section 5 — Core Business Workflows & State Machines

5.1 Order Placement & Server-Authoritative Pipeline:

- **Step 1 — Cart Checkout:** Customer initiates checkout. Frontend packages selected product IDs, quantities, address, and coupon code.
- **Step 2 — API Ingestion:** Request hits POST /api/orders on Cloud Run with Bearer Firebase ID Token. Zod validates payload format.
- **Step 3 — Price Lookup & Inventory Lock:** Backend queries Firestore for active product documents, verifies stock availability, and computes authoritative subtotals.
- **Step 4 — Coupon Application:** If coupon provided, verifies validity window and user usage history before applying discount.
- **Step 5 — Order Creation:** Generates 6-digit random delivery OTP, writes order record with paymentStatus: 'Pending (COD)' or 'Pending UTR Verification', and returns order ID.

5.2 Delivery Fleet Dispatch & Fulfillment State Machine:

- **1. Assigned (assigned):** Admin assigns order to a delivery partner via Smart Load Balancing (matching zone and lowest active orders).
- **2. Accepted (accepted):** Rider receives push notification / queue update and accepts assignment. (Status: *Processing*).
- **3. Reached Store (reached_store):** Rider arrives at vendor hub to collect packaged garments.
- **4. Picked Up (picked_up):** Rider confirms package pickup from merchant. (Status: *Shipped*).
- **5. Out for Delivery (out_for_delivery):** Rider departs towards customer address using built-in Google Maps navigation.
- **6. Reached Customer (reached_customer):** Rider arrives at customer doorstep, triggers phone call if necessary.
- **7. Delivered (delivered):** Customer provides 6-digit OTP. For COD orders, cash is collected and recorded. Status updates to *Delivered*.

5.3 Payment Verification Workflow (Manual UPI & COD):

- **Manual BharatPe UPI VPA:** Dynamic QR code generated with VPA (BHARATPE.8E0Q0V9A0U38126@fbpe, Payee: Anshu Kumari). Customer inputs 12-digit bank UTR reference number upon payment.
- **UTR Anti-Tampering Engine:** Submissions are recorded in /payments. Duplicate UTRs are rejected immediately. Status defaults to Pending UTR Verification.
- **Admin Reconciliation:** Admin confirms bank account receipt and flags order as Paid via PATCH /api/admin/orders/:id/verify-payment.
- **Cash on Delivery (COD):** Available for orders up to Rs. 5,000. Rider collects exact cash upon delivery handoff and system marks COD as collected.

Section 6 — Progressive Web App (PWA) & Caching Architecture

The client application implements a Service Worker (sw.js v6.0) utilizing tailored caching strategies to balance ultra-fast load times with live data consistency:

Traffic / Asset Category	Caching Strategy	Behavior & Rationale
HTML Navigation Pages (*.html)	Network-First with Offline Fallback	Fetches fresh HTML from CDN; serves cached version or offline.html if offline.
Static Core Assets (CSS, Icons, Brand Logos)	Stale-While-Revalidate	Serves cached assets instantly while fetching background updates for next visit.
Delivery & Admin Portals (/delivery/*, /admin/*)	Strict Dynamic Bypass	Completely bypasses Service Worker cache to guarantee 100% real-time Firestore synchronization.
Cloud Run API Traffic (*.run.app)	Strict Dynamic Bypass	Direct network pass-through to ensure accurate pricing calculations and atomic order creations.

Section 7 — Infrastructure Deployment & Production Endpoints

Resource / Environment	Production URL / Identifier	Provider & Region
Primary Custom Domain	https://nariniketan.shop & https://www.nariniketan.shop	Google Edge CDN / Hostinger DNS
Firebase Hosting Origin	https://nari-niketan.web.app	Google Cloud Firebase CDN
Backend Cloud Run API	https://nari-niketan-api-997712460310.asia-south1.run.app	Google Cloud Run (asia-south1, Mumbai)
Firestore NoSQL DB	nari-niketan (default instance)	Google Cloud Firestore (asia-south1)
Firebase Storage Bucket	gs://nari-niketan.appspot.com	Google Cloud Storage (Multi-region asia)